

Year 7 · Chapter 8 · Area and volume: answers (teacher copy)

8.01 Units of measurement (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: 3.2 km
- Ex 1b: 42 g
- Ex 2a: 8300 mL
- Ex 2b: 4 h
- Ex 3a: 54 000 s
- Ex 3b: 6 307 200 min (B)

Easy (drill)

- | | | |
|---------------|--------------|------------|
| 1. 22 cm | 11. 3 300 mg | 21. 11 h |
| 2. 5 mm | 12. 31 000 m | 22. 72 h |
| 3. 36 m | 13. 180 s | 23. 96 h |
| 4. 280 L | 14. 2 h | 24. 3 h |
| 5. 48 000 mm | 15. 120 min | 25. 480 s |
| 6. 0.13 L | 16. 420 min | 26. 6 h |
| 7. 23 000 mm | 17. 300 min | 27. 420 s |
| 8. 0.15 m | 18. 144 h | 28. 60 min |
| 9. 0.45 kL | 19. 600 s | |
| 10. 53 000 mL | 20. 360 s | |

Medium (drill)

- | | | |
|--------------|--------------|-------------|
| 1. 570 cm | 9. 1.8 km | 17. 285 min |
| 2. 51 000 mg | 10. 8.2 kg | 18. 555 min |
| 3. 3 000 mm | 11. 910 mm | 19. 8.75 h |
| 4. 830 cm | 12. 1 700 m | 20. 4.25 h |
| 5. 55 m | 13. 9 600 cm | 21. 9.5 h |
| 6. 180 cm | 14. 4.6 kg | 22. 42 h |
| 7. 0.4 cm | 15. 50 kL | |
| 8. 56 000 L | 16. 9 100 g | |

Challenging

- 6 307 200 min
- 4.2 kg
- 7.2 L

Extension

- about 36 792 000
- 11.6 days

Exit ticket

- Level 1: 450 cm
- Level 2: 2.75 kg
- Level 3: 210 min

8.02 Perimeter (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: 52 m
- Ex 1b: 36 mm
- Ex 2a: 78 cm
- Ex 2b: 650 cm
- Ex 3a: 128 mm
- Ex 3b: 11.5 cm

Easy (drill)

- | | | |
|---------|------------|------------|
| 1. 34 m | 10. 16 m | 19. 66 cm |
| 2. 96 m | 11. 52 m | 20. 60 cm |
| 3. 10 m | 12. 52 m | 21. 128 cm |
| 4. 14 m | 13. 40 cm | 22. 48 cm |
| 5. 60 m | 14. 85 cm | 23. 9 cm |
| 6. 36 m | 15. 192 cm | 24. 48 cm |
| 7. 74 m | 16. 52 cm | 25. 102 cm |
| 8. 6 m | 17. 96 cm | 26. 36 cm |
| 9. 72 m | 18. 24 cm | 27. 88 cm |

28. 54 cm

Medium (drill)

- | | | |
|----------|--------------------------|--------------------------|
| 1. 13 m | 14. 5 m | 21. 18.8 m (or 1880 cm) |
| 2. 9 m | 15. 22 m | 22. 8.46 m (or 846 cm) |
| 3. 4 m | 16. 2.26 m (or 226 cm) | 23. 18.36 m (or 1836 cm) |
| 4. 3 m | 17. 6.92 m (or 692 cm) | 24. 17.28 m (or 1728 cm) |
| 5. 3 m | 18. 17.42 m (or 1742 cm) | 25. 10.86 m (or 1086 cm) |
| 6. 5 m | 19. 12.24 m (or 1224 cm) | 26. 5.7 m (or 570 cm) |
| 7. 3 m | 20. 11.64 m (or 1164 cm) | 27. 18.4 m (or 1840 cm) |
| 8. 4 m | | |
| 9. 5 m | | |
| 10. 2 m | | |
| 11. 10 m | | |
| 12. 23 m | | |
| 13. 14 m | | |

Challenging

- \$2597.00
- 9.5 cm
- 5 (1×9, 2×8, 3×7, 4×6, 5×5)

Extension

- 10 cm by 15 cm
- 14 cm

Exit ticket

- Level 1: 38 cm
- Level 2: 28.5 m
- Level 3: 15 mm

8.03 Circumference of a circle (booklet p. 13)

WE DO examples (teacher)

- Ex 1a: 6.3 cm
- Ex 1b: 62.8 mm
- Ex 2a: 44.0 cm
- Ex 2b: 175.9 mm
- Ex 3a: 207.3 m
- Ex 3b: 15.9 cm

Easy (drill)

- | | | |
|-------------|-------------|-------------|
| 1. 100.5 cm | 11. 31.4 cm | 21. 18.8 m |
| 2. 44.0 cm | 12. 15.7 cm | 22. 75.4 m |
| 3. 28.3 cm | 13. 37.7 m | 23. 88.0 m |
| 4. 59.7 cm | 14. 125.7 m | 24. 44.0 m |
| 5. 88.0 cm | 15. 94.2 m | 25. 69.1 m |
| 6. 18.8 cm | 16. 100.5 m | 26. 6.3 m |
| 7. 69.1 cm | 17. 31.4 m | 27. 131.9 m |
| 8. 34.6 cm | 18. 50.3 m | 28. 81.7 m |
| 9. 110.0 cm | 19. 12.6 m | |
| 10. 40.8 cm | 20. 56.5 m | |

Medium (drill)

- | | | |
|------------|-------------|-------------|
| 1. 46.8 cm | 9. 11.5 cm | 17. 28.3 cm |
| 2. 19.1 cm | 10. 29.6 cm | 18. 66.8 cm |
| 3. 40.7 cm | 11. 28.6 cm | 19. 41.1 cm |
| 4. 37.9 cm | 12. 22.9 cm | 20. 12.9 cm |
| 5. 17.2 cm | 13. 51.4 cm | 21. 30.8 cm |
| 6. 12.7 cm | 14. 25.7 cm | 22. 59.1 cm |
| 7. 22.6 cm | 15. 43.7 cm | 23. 38.6 cm |
| 8. 29.3 cm | 16. 18.0 cm | 24. 64.3 cm |

Challenging

- 207.3 m

- 15.9 cm
- 30.8 cm

Extension

- 36.0 m
- 401.1 m

Exit ticket

- Level 1: 16 cm
- Level 2: 28.3 m
- Level 3: 22.0 cm

8.04 Area units (booklet p. 19)

WE DO examples (teacher)

- Ex 1a: a 1 cm square is 10 mm by 10 mm
- Ex 1b: 91 000
- Ex 2a: 250
- Ex 2b: 17.5
- Ex 3a: 0.24
- Ex 3b: 8 600 000

Easy (drill)

- | | | |
|---------------------------|-----------------------------|-----------------------------|
| 1. 5 400 cm ² | 11. 60 km ² | 20. 100 000 cm ² |
| 2. 49 km ² | 12. 4 100 cm ² | 21. 240 000 cm ² |
| 3. 4 300 mm ² | 13. 300 000 cm ² | 22. 170 000 cm ² |
| 4. 2 900 ha | 14. 10 000 cm ² | 23. 290 000 cm ² |
| 5. 27 cm ² | 15. 200 mm ² | 24. 1 100 mm ² |
| 6. 3 300 mm ² | 16. 2 400 mm ² | 25. 1 900 mm ² |
| 7. 26 ha | 17. 270 000 cm ² | 26. 190 000 cm ² |
| 8. 12 000 cm ² | 18. 280 000 cm ² | 27. 600 mm ² |
| 9. 41 000 cm ² | 19. 400 mm ² | 28. 260 000 cm ² |
| 10. 3 100 mm ² | | |

Medium (drill)

- | | | |
|---------------------------|-----------------------------|---------------------|
| 1. 400 mm ² | 9. 670 000 m ² | 17. m ² |
| 2. 7 600 ha | 10. 980 000 m ² | 18. km ² |
| 3. 790 mm ² | 11. 70 mm ² | 19. km ² |
| 4. 62 000 cm ² | 12. 75 m ² | 20. cm ² |
| 5. 530 mm ² | 13. 320 000 cm ² | 21. cm ² |
| 6. 37 ha | 14. 1 800 ha | 22. m ² |
| 7. 21 000 m ² | 15. 6 m ² | |
| 8. 60 000 cm ² | 16. 6 600 ha | |

Challenging

- mm² or cm²; m²; ha; km²
- 3.2 m² (30 000 cm² = 3 m²)
- 150 tiles

Extension

- 1 000 000 mm²
- 0.71 ha

Exit ticket

- Level 1: 500 mm²
- Level 2: 3 m²
- Level 3: 1 500 000 mm²

8.05 Area of squares and rectangles

(booklet p. 23)

WE DO examples (teacher)

- Ex 1a: 16 m²
- Ex 1b: 24 cm²
- Ex 2a: 162 m²
- Ex 2b: 1.5 m²

- Ex 3a: 9 cm
- Ex 3b: 6 m

Easy (drill)

- | | | |
|-----------------------|-------------------------|-------------------------|
| 1. 112 m ² | 11. 72 m ² | 21. 324 cm ² |
| 2. 4 m ² | 12. 22 m ² | 22. 196 cm ² |
| 3. 40 m ² | 13. 361 cm ² | 23. 36 cm ² |
| 4. 68 m ² | 14. 4 cm ² | 24. 16 cm ² |
| 5. 256 m ² | 15. 49 cm ² | 25. 169 cm ² |
| 6. 119 m ² | 16. 100 cm ² | 26. 289 cm ² |
| 7. 40 m ² | 17. 225 cm ² | 27. 25 cm ² |
| 8. 70 m ² | 18. 64 cm ² | 28. 9 cm ² |
| 9. 6 m ² | 19. 1 cm ² | |
| 10. 36 m ² | 20. 81 cm ² | |

Medium (drill)

- | | | |
|----------|--------------------------|--------------------------|
| 1. 8 m | 11. 3 m | 20. 11.4 cm ² |
| 2. 15 m | 12. 5 m | 21. 27 cm ² |
| 3. 16 m | 13. 15 m | 22. 35.4 cm ² |
| 4. 2 m | 14. 10 m | 23. 62 cm ² |
| 5. 6 m | 15. 9 m | 24. 40.5 cm ² |
| 6. 19 m | 16. 36 cm ² | 25. 64.8 cm ² |
| 7. 13 m | 17. 22.8 cm ² | 26. 89.1 cm ² |
| 8. 14 m | 18. 21.6 cm ² | 27. 45.5 cm ² |
| 9. 14 m | 19. 28.2 cm ² | |
| 10. 18 m | | |

Challenging

- \$936
- 108 tiles
- 8 cm by 5 cm

Extension

- 24 m²
- 6 cm by 6 cm (36 cm²)

Exit ticket

- Level 1: 54 cm²
- Level 2: 6.25 m²
- Level 3: 12 m

8.06 Area of a triangle (booklet p. 29)

WE DO examples (teacher)

- Ex 1a: 360 mm²
- Ex 1b: 112 mm²
- Ex 2a: 5.2 cm²
- Ex 2b: 187 km²
- Ex 3a: 8 cm
- Ex 3b: two copies of the triangle make a rectangle (or parallelogram)

Easy (drill)

- | | | |
|------------------------|--------------------------|--------------------------|
| 1. 28 cm ² | 10. 152 m ² | 19. 184 m ² |
| 2. 54 cm ² | 11. 150 m ² | 20. 252 m ² |
| 3. 14 cm ² | 12. 49.5 m ² | 21. 237.5 m ² |
| 4. 152 cm ² | 13. 198 m ² | 22. 170 m ² |
| 5. 10 cm ² | 14. 100 m ² | 23. 90 m ² |
| 6. 36 cm ² | 15. 28 m ² | 24. 237.5 m ² |
| 7. 14 cm ² | 16. 253 m ² | 25. 65 m ² |
| 8. 33 cm ² | 17. 137.5 m ² | |
| 9. 70 cm ² | 18. 55 m ² | |

Medium (drill)

- | | | |
|----------|----------|----------|
| 1. 8 cm | 3. 13 cm | 5. 8 cm |
| 2. 16 cm | 4. 3 cm | 6. 17 cm |

- | | | |
|-----------|-------------------------|-------------------------|
| 7. 20 cm | 13. 1.1 m ² | 19. 7.2 m ² |
| 8. 4 cm | 14. 26.8 m ² | 20. 17.6 m ² |
| 9. 11 cm | 15. 37 m ² | 21. 35.2 m ² |
| 10. 14 cm | 16. 32.8 m ² | 22. 2.8 m ² |
| 11. 4 cm | 17. 13.8 m ² | 23. 10.8 m ² |
| 12. 2 cm | 18. 5.2 m ² | 24. 12.9 m ² |

Challenging

- 8 m²
- 4 bags
- 92 cm²

Extension

- 27 m²
- 24 square units

Exit ticket

- Level 1: 35 m²
- Level 2: 8 cm²
- Level 3: 8 cm

8.07 Area of a parallelogram (booklet p. 35)

WE DO examples (teacher)

- Ex 1a: 120 m²
- Ex 1b: 828 cm²
- Ex 2a: 68 mm²
- Ex 2b: cutting off a triangle and moving it makes a rectangle
- Ex 3a: 6 cm
- Ex 3b: 50 cm²

Easy (drill)

- | | | |
|-----------------------|-------------------------|-------------------------|
| 1. 120 m ² | 10. 132 cm ² | 19. 90 cm ² |
| 2. 30 m ² | 11. 245 cm ² | 20. 192 cm ² |
| 3. 54 m ² | 12. 348 cm ² | 21. 130 cm ² |
| 4. 70 m ² | 13. 660 cm ² | 22. 189 cm ² |
| 5. 18 m ² | 14. 612 cm ² | 23. 180 cm ² |
| 6. 12 m ² | 15. 550 cm ² | 24. 65 cm ² |
| 7. 34 m ² | 16. 324 cm ² | 25. 459 cm ² |
| 8. 126 m ² | 17. 372 cm ² | |
| 9. 144 m ² | 18. 117 cm ² | |

Medium (drill)

- | | | |
|---------|--------------------------|--------------------------|
| 1. 10 m | 10. 2 m | 18. 35.67 m ² |
| 2. 12 m | 11. 4 m | 19. 34.79 m ² |
| 3. 10 m | 12. 12 m | 20. 10.54 m ² |
| 4. 6 m | 13. 13.44 m ² | 21. 8.16 m ² |
| 5. 8 m | 14. 11.75 m ² | 22. 28 m ² |
| 6. 17 m | 15. 14.56 m ² | 23. 30.4 m ² |
| 7. 14 m | 16. 15.18 m ² | 24. 12.39 m ² |
| 8. 5 m | 17. 5.22 m ² | |
| 9. 6 m | | |

Challenging

- \$882
- e.g. 6 × 4 and 8 × 3
- 80 cm²

Extension

- 32 m²
- 12 cm

Exit ticket

- Level 1: 36 cm²
- Level 2: 4 m²
- Level 3: 12.5 m

8.08 Area of composite shapes (booklet p. 41)

WE DO examples (teacher)

- Ex 1a: 23 cm²
- Ex 1b: 71 m²
- Ex 2a: 30 m²
- Ex 2b: 360 m²
- Ex 3a: 6 L
- Ex 3b: 37.5 mm²

Easy (drill)

- | | | |
|------------------------|------------------------|-----------------------|
| 1. 96 cm ² | 8. 51 m ² | 15. 94 m ² |
| 2. 135 cm ² | 9. 90 m ² | 16. 62 m ² |
| 3. 101 cm ² | 10. 38 m ² | 17. 59 m ² |
| 4. 70 cm ² | 11. 126 m ² | 18. 63 m ² |
| 5. 170 cm ² | 12. 20 m ² | 19. 63 m ² |
| 6. 111 cm ² | 13. 74 m ² | 20. 92 m ² |
| 7. 66 m ² | 14. 56 m ² | |

Medium (drill)

- 30 m²
- 68 m²
- 99 m²
- 110 m²
- 65 m²
- 110 m²
- 60 (units²)
- 74 (units²)
- 68 (units²)

Challenging

- 6 L
- 630 cm²
- 21 m²

Extension

- 57 m²
- 45 cm²

Exit ticket

- Level 1: 46 cm²
- Level 2: 22 m²
- Level 3: 3 L

8.09 Prisms, cross-sections and views

(booklet p. 47)

WE DO examples (teacher)

- Ex 1a: yes; rectangle
- Ex 1b: yes; triangle
- Ex 2a: L-shape
- Ex 2b: no: the cross-section changes and the ends are not the same
- Ex 3a: front: L-shape; left: rectangle; top: rectangle
- Ex 3b: right: rectangle; front: pentagon (house shape); top: rectangle with a line

Easy (drill)

- | | | |
|--------------|----------------------|---------------------|
| 1. circle | 7. square | 13. F 7, V 10, E 15 |
| 2. hexagon | 8. rectangle | 14. F 5, V 5, E 8 |
| 3. hexagon | 9. pentagon | 15. F 6, V 8, E 12 |
| 4. octagon | 10. F 10, V 16, E 24 | 16. F 4, V 4, E 6 |
| 5. circle | 11. F 8, V 12, E 18 | 17. F 5, V 6, E 9 |
| 6. rectangle | 12. F 6, V 8, E 12 | |

Medium (drill)

- | | | |
|--------|-------|--------|
| 1. yes | 2. no | 3. yes |
|--------|-------|--------|

- | | | |
|------------------------|--------|---------|
| 4. no (curved surface) | 7. yes | 10. no |
| 5. no | 8. yes | 11. yes |
| 6. no | 9. yes | 12. yes |

Challenging

1. T-shape; rectangle; rectangle
2. a cylinder

Extension

1. a square pyramid
2. three squares in a row of heights 1, 2, 3 (or one tall column, depending on view)

Exit ticket

1. Level 1: No: it has curved sides (it is like a prism with a circle cross-section)
2. Level 2: hexagon
3. Level 3: a rectangle with a line along the middle

8.10 Volume units (booklet p. 53)

WE DO examples (teacher)

1. Ex 1a: a 1 cm cube is $10 \times 10 \times 10$ mm
2. Ex 1b: 20 000
3. Ex 2a: 500 000
4. Ex 2b: 7.5
5. Ex 3a: 0.00023
6. Ex 3b: 60 cubes

Easy (drill)

- | | | |
|-------------------------------|-----------------------------|-------------------------|
| 1. 27 000 mm ³ | 10. 3 cm ³ | 19. 334 mL |
| 2. 38 000 000 cm ³ | 11. 540 000 cm ³ | 20. 538 cm ³ |
| 3. 0.16 cm ³ | 12. 6 cm ³ | 21. 9.5 L |
| 4. 2.3 m ³ | 13. 640 cm ³ | 22. 642 mL |
| 5. 9 000 mm ³ | 14. 466 mL | 23. 8.14 L |
| 6. 0.42 m ³ | 15. 7.33 L | 24. 0.62 L |
| 7. 14 cm ³ | 16. 1.32 L | 25. 4.38 L |
| 8. 37 000 mm ³ | 17. 368 mL | 26. 559 cm ³ |
| 9. 53 000 000 cm ³ | 18. 8.48 L | 27. 0.95 L |
| | | 28. 6.49 L |

Medium (drill)

- | | | |
|--------------------------|-------------------------------|-------------------------|
| 1. 7 600 mm ³ | 8. 5 200 000 cm ³ | 15. 59 m ³ |
| 2. 3 800 mm ³ | 9. 31 cm ³ | 16. 2.8 cm ³ |
| 3. 8 000 mm ³ | 10. 4 100 000 cm ³ | 17. 43 000 L |
| 4. 4.7 m ³ | 11. 8 900 mm ³ | 18. 61 000 L |
| 5. 7.7 cm ³ | 12. 46 cm ³ | 19. 3 000 L |
| 6. 0.2 m ³ | 13. 4 500 000 cm ³ | 20. 15 m ³ |
| 7. 8.7 cm ³ | 14. 4.9 cm ³ | 21. 47 m ³ |
| | | 22. 77 m ³ |

Challenging

1. 60
2. 30
3. 15 000 cm³, 0.02 m³, 25 000 000 mm³

Extension

1. 350 000 mm³
2. 1000

Exit ticket

1. Level 1: 6000 mm³
2. Level 2: 2.5 m³
3. Level 3: 1.5 cm³

8.11 Volume of a rectangular prism (booklet p. 57)

WE DO examples (teacher)

1. Ex 1a: 80 m³
2. Ex 1b: 320 cm³
3. Ex 2a: 672 mm³
4. Ex 2b: 3000 cm³
5. Ex 3a: 4 cm
6. Ex 3b: 5 cm

Easy (drill)

- | | | |
|------------------------|--------------------------|--------------------------|
| 1. 60 cm ³ | 10. 1728 cm ³ | 19. 216 m ³ |
| 2. 288 cm ³ | 11. 1 cm ³ | 20. 8 cm ³ |
| 3. 198 cm ³ | 12. 216 cm ³ | 21. 512 m ³ |
| 4. 108 cm ³ | 13. 729 m ³ | 22. 1000 cm ³ |
| 5. 108 cm ³ | 14. 1331 cm ³ | 23. 64 cm ³ |
| 6. 630 cm ³ | 15. 125 m ³ | 24. 8 m ³ |
| 7. 162 cm ³ | 16. 343 cm ³ | 25. 512 cm ³ |
| 8. 252 cm ³ | 17. 1000 m ³ | |
| 9. 32 cm ³ | 18. 125 cm ³ | |

Medium (drill)

- | | | |
|---------|------------------------|--------------------------|
| 1. 5 cm | 10. 7 cm | 18. 51.6 m ³ |
| 2. 6 cm | 11. 7 cm | 19. 116 m ³ |
| 3. 3 cm | 12. 2 cm | 20. 114 m ³ |
| 4. 7 cm | 13. 168 m ³ | 21. 64.8 m ³ |
| 5. 7 cm | 14. 55 m ³ | 22. 38.4 m ³ |
| 6. 8 cm | 15. 35 m ³ | 23. 156.8 m ³ |
| 7. 4 cm | 16. 106 m ³ | 24. 65 m ³ |
| 8. 6 cm | 17. 32 m ³ | |
| 9. 9 cm | | |

Challenging

1. 74.88 m³
2. 72 cubes
3. 0.96 m³

Extension

1. 540 cm³
2. 108 s

Exit ticket

1. Level 1: 60 cm³
2. Level 2: 216 m³
3. Level 3: 6 cm

8.12 Volume of a prism (booklet p. 63)

WE DO examples (teacher)

1. Ex 1a: 554.472 cm³
2. Ex 1b: 2200 mm³
3. Ex 2a: 570 m³
4. Ex 2b: 280 cm³
5. Ex 3a: 120 cm³
6. Ex 3b: 30 m²

Easy (drill)

- | | | |
|------------------------|-------------------------|-------------------------|
| 1. 696 cm ³ | 8. 171 cm ³ | 15. 216 cm ³ |
| 2. 40 cm ³ | 9. 153 cm ³ | 16. 210 cm ³ |
| 3. 33 cm ³ | 10. 324 cm ³ | 17. 180 cm ³ |
| 4. 440 cm ³ | 11. 462 cm ³ | 18. 840 cm ³ |
| 5. 286 cm ³ | 12. 9 cm ³ | 19. 126 cm ³ |
| 6. 56 cm ³ | 13. 420 cm ³ | 20. 48 cm ³ |
| 7. 84 cm ³ | 14. 450 cm ³ | 21. 168 cm ³ |

Medium (drill)

- | | | |
|---------|-----------------------|-------------------------|
| 1. 8 m | 8. 3 m | 14. 136 m^3 |
| 2. 14 m | 9. 9 m | 15. 264 m^3 |
| 3. 9 m | 10. 14 m | 16. 137.5 m^3 |
| 4. 7 m | 11. 4 m | 17. 126 m^3 |
| 5. 4 m | 12. 2 m | 18. 56 m^3 |
| 6. 5 m | 13. 360 m^3 | |
| 7. 13 m | | |

Challenging

1. 5.4 m^3
2. 0.6 m^3
3. 30 m^2

Extension

1. 1600 m^3
2. $375\,000 \text{ L}$

Exit ticket

1. Level 1: 90 cm^3
2. Level 2: 90 m^3
3. Level 3: 8 cm

8.13 Capacity (booklet p. 69)**WE DO examples (teacher)**

1. Ex 1a: 6 L
2. Ex 1b: 15 cm (D)
3. Ex 2a: 3500 L
4. Ex 2b: 80 000 L
5. Ex 3a: 2 000 000 000 ML
6. Ex 3b: 8 cups

Easy (drill)

- | | | |
|--------------|-------------------------|-------------------------|
| 1. 0.05 kL | 11. 17 L | 20. 1000 cm^3 |
| 2. 1.6 L | 12. 18 L | 21. 1700 cm^3 |
| 3. 5 000 L | 13. 4300 cm^3 | 22. 2800 cm^3 |
| 4. 19 000 mL | 14. 800 cm^3 | 23. 3.2 L |
| 5. 58 000 L | 15. 3.9 L | 24. 4.1 L |
| 6. 300 mL | 16. 1200 cm^3 | 25. 1.8 L |
| 7. 57 000 L | 17. 3.5 L | 26. 4400 cm^3 |
| 8. 580 mL | 18. 3800 cm^3 | 27. 3.1 L |
| 9. 420 mL | 19. 4800 cm^3 | 28. 2300 cm^3 |
| 10. 3 900 L | | |

Medium (drill)

- | | | |
|-----------|----------|-------------|
| 1. 10 L | 7. 9 L | 13. 18 cups |
| 2. 12.5 L | 8. 24 L | 14. 24 cups |
| 3. 62.5 L | 9. 12 L | 15. 36 cups |
| 4. 12 L | 10. 3 L | 16. 45 cups |
| 5. 60 L | 11. 25 L | 17. 10 cups |
| 6. 16 L | 12. 20 L | 18. 30 cups |

Challenging

1. 15 cm
2. 2 000 000 000 ML
3. 63 L

Extension

1. 50 mm
2. $1\frac{1}{3}$ days

Exit ticket

1. Level 1: 2.5 L
2. Level 2: 5 L
3. Level 3: 600 L